WAYGROUND formerly Quizizz **Worksheets**

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ITA03 MOBILE COMPUTING CO

Class: Slot A-Mobile COmputing

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1. A network engineer is designing a new cellular system for a densely populated urban area. The primary challenge is the extremely high user density, leading to significant co-channel interference. To maximize spectral efficiency and user capacity, the engineer decides to implement a multiple access scheme that allows multiple users to share the same frequency channel at the same time by separating them based on their unique spatial signatures. This technique requires advanced antenna arrays and signal processing to steer beams towards specific users. Which of the following multiple access techniques is the engineer primarily employing?

a) Time Division Multiple Access (TDMA)

b) Code Division Multiple Access (CDMA)

c) Space Division Multiple Access (SDMA)

d) Frequency Division Multiple Access (FDMA)

2. A telecommunications analyst is comparing the fundamental differences between mobile communication and mobile computing. While mobile communication focuses on the transmission of voice and data over wireless networks, mobile computing encompasses a broader scope. An executive argues that the primary distinction lies in the device's capability to process data locally rather than just relaying it. Considering the core definitions and evolution of these fields, which of the following scenarios best illustrates the shift from a pure mobile communication paradigm to a mobile computing paradigm?

a) A GSM handset authenticating with the network using a SIM card.

b) A legacy pager receiving a one-way text message.

c) A smartphone running a navigation app that processes GPS data locally to provide real-time traffic updates.

d) A traditional cellular phone making a circuit-switched voice call to a landline.

3. A radio network planner is evaluating the performance of different duplexing methods. In Frequency Division Duplexing (FDD), uplink and downlink transmissions occur on separate frequency bands, requiring a guard band to prevent interference. In Time Division Duplexing (TDD), both transmissions occur on the same frequency band but in different time slots. The planner is tasked with deploying a system in a region with highly asymmetric traffic patterns, where the downlink data demand (e.g., video streaming) far exceeds the uplink demand. Which of the following is the most significant advantage of TDD over FDD in this specific scenario?

a) TDD requires simpler and less expensive RF filters.

b) TDD allows for dynamic allocation of time slots to better match the asymmetric traffic load.

c) TDD provides better coverage for a given transmit power.

d) TDD is less susceptible to multipath fading.

4. An engineer is analyzing the capacity of a cellular system that uses an FDMA scheme. The total available bandwidth is 25 MHz, and each voice channel requires a bandwidth of 25 kHz, including guard bands. The engineer calculates the total number of available channels. However, to reduce co-channel interference, the network employs a frequency reuse factor of 1/7, meaning the available channels are divided among 7 cells in a cluster. What is the total number of voice channels available per cell in this specific system configuration?

a) 142.85 channels (approximately 143)

b) 7000 channels

c) 1000 channels

d) 25 channels

5. A network designer is choosing between TDMA and FDMA for a new satellite communication system. The designer is concerned about the non-linear effects of the High-Power Amplifier (HPA) on the satellite transponder. In FDMA, multiple carriers are amplified simultaneously, which forces the HPA to operate in a back-off region to avoid intermodulation distortion, thus reducing its efficiency. In TDMA, only one carrier uses the transponder at a time, allowing the HPA to operate at or near saturation. Considering this, what is the primary advantage of choosing TDMA over FDMA in this context?

a) TDMA provides a lower delay in data transmission.
b) TDMA allows for simpler synchronization between the earth station and the satellite.
c) TDMA eliminates the need for a guard band between channels.
d) TDMA allows the satellite's power amplifier to operate more efficiently with less intermodulation distortion.

6. A research team is developing a new wireless standard for the Internet of Things (IoT) where many devices need to transmit small amounts of data sporadically. The team is considering CDMA as a potential access method. One of the key challenges in a CDMA system is the "near-far" problem, where a nearby high-power transmitter can drown out the signal from a distant low-power transmitter. To mitigate this, the system employs a sophisticated power control mechanism. What is the primary objective of this power control mechanism in a CDMA system?

a) To ensure that all mobile devices transmit at their maximum power.
b) To ensure that the signal from each user arrives at the base station with the same power level.
c) To allocate different frequency bands to mobile devices based on their distance from the base station.
d) To reduce the battery consumption of mobile devices.

7. A telecommunications historian is documenting the evolution of mobile networks from 1G to 5G. They note that each generation introduced significant technological leaps. The transition from 1G (analog) to 2G (digital) was revolutionary, primarily due to the introduction of digital modulation and multiple access techniques. Which of the following was the most significant advantage of 2G systems like GSM over 1G systems like AMPS, fundamentally changing the way cellular networks operated and enabling future data services?

a) The use of circuit-switched technology for voice calls.
b) The introduction of Frequency Division Duplexing (FDD).
c) The introduction of digital encryption and support for SMS and data services.
d) The use of a cellular architecture with handover capabilities.

8. A network operator is upgrading its 3G UMTS network to support higher data rates. They are considering the implementation of High-Speed Downlink Packet Access (HSDPA). HSDPA introduces several new features to the UMTS air interface, including Adaptive Modulation and Coding (AMC) and fast scheduling. AMC allows the system to dynamically change the modulation scheme (e.g., QPSK, 16-QAM) and coding rate based on the channel quality reported by the mobile device. What is the primary benefit of using AMC in HSDPA?

a) It simplifies the design of the mobile receiver.
b) It allows the network to adapt to varying radio conditions, optimizing throughput for each user.
c) It reduces the need for handovers between cells.
d) It increases the coverage area of each cell.

9. A mobile network operator is analyzing the performance of its GSM network. A subscriber is traveling from a city center to a suburb, crossing several cell boundaries. The network must ensure the call is not dropped. The process involves the Mobile Switching Center (MSC), Base Station Controllers (BSCs), and Base Transceiver Stations (BTSs). In GSM, there are two main types of handover: intra-BSC and inter-BSC (which may be intra-MSC or inter-MSC). The subscriber is currently in a call, connected to a BTS managed by BSC-1, and is moving towards an area served by BTS-2, which is managed by BSC-2. Both BSCs are connected to the same MSC. Which sequence of events correctly describes the network elements involved in performing this handover?

a) BSC-1 informs MSC of the need for a handover, MSC coordinates with BSC-2, and then instructs BSC-1 to execute the handover.
b) The MS initiates the handover and communicates directly with the new BTS to establish a connection.
c) BSC-1 makes the handover decision and controls the entire process, informing the MS to switch to BSC-2's frequency.
d) The MSC bypasses the BSCs and directly controls the handover between the two BTSs.

10. A network engineer is configuring the location update procedure for a GSM network. The network is divided into several Location Areas (LAs), each identified by a Location Area Identity (LAI). When a mobile subscriber (MS) moves from one LA to another, it must perform a Location Update to inform the network of its new location. However, the MS also performs a periodic location update even if it hasn't moved to a new LA. What is the primary purpose of this periodic location update, and what network element stores the subscriber's current LAI?

a) To update the MS's service subscription; stored in the HLR.
b) To update the MS's battery status; stored in the BSC.
c) To confirm the MS's identity; stored in the EIR.
d) To ensure the network is aware of the MS's availability and location in case of a power-off or out-of-coverage event; stored in the VLR.

11. A systems analyst is comparing the architectural differences between GSM and GPRS. GSM is fundamentally a circuit-switched network designed for voice, while GPRS is a packet-switched overlay network designed for data. An engineer argues that GPRS is more efficient for bursty data applications because it does not permanently dedicate resources for a single user. The analyst counters by pointing out a significant architectural addition in GPRS. Which of the following is the core network element introduced in GPRS that is responsible for routing and forwarding data packets to and from the mobile users, essentially acting as a gateway to external packet data networks like the Internet?

a) Serving GPRS Support Node (SGSN) and Gateway GPRS Support Node (GGSN)
b) Base Station Controller (BSC)
c) Home Location Register (HLR)
d) Mobile Switching Center (MSC)

12. A technician is studying different standards used in mobile communications. They come across the Digital Enhanced Cordless Telecommunications (DECT) standard. DECT is widely used for cordless phones and wireless office systems. The technician notes that unlike GSM or UMTS, which are designed for wide-area coverage, DECT is designed for short-range, low-mobility applications. Considering its key characteristics and applications, which of the following statements accurately differentiates DECT from a typical cellular network standard like GSM?

a) DECT operates in a licensed spectrum band, whereas GSM operates in an unlicensed band.
b) DECT is a packet-switched technology only, while GSM is circuit-switched only.
c) DECT primarily uses a Time Division Multiple Access (TDMA) scheme with a fixed frame structure for cordless telephony.
d) DECT supports high-speed vehicular handover, unlike GSM which is only for pedestrian users.

13. A network designer is evaluating the security protocols in different generations of mobile networks. One of the primary security goals is to prevent a rogue base station from impersonating a legitimate network and tricking a mobile device into connecting to it. In GSM, the network authenticates the mobile device, but the mobile device does not authenticate the network. This vulnerability was addressed in UMTS with a mutual authentication procedure. What mechanism in UMTS enables the mobile device to authenticate the network, thereby mitigating the "false base station" attack?

a) The use of Frequency Division Duplexing (FDD) for the air interface.
b) The introduction of the Authentication Token (AUTN) in the authentication process.
c) The use of a longer Temporary Mobile Subscriber Identity (TMSI).
d) The use of a stronger encryption algorithm (A5/3).

14. A mobile network operator is planning its transition from 4G LTE to 5G NR (New Radio). One of the key differences between the two is the approach to the core network architecture. The 5G core network is designed to be service-based and more flexible, supporting network slicing. Which of the following concepts is central to the 5G core network architecture and is a significant departure from the Evolved Packet Core (EPC) used in LTE?

a) A service-based architecture (SBA) where network functions communicate via standardized service-based interfaces.
b) A dedicated hardware-based approach for all network functions.
c) A single, monolithic Mobility Management Entity (MME).
d) The continued use of circuit-switched fallback (CSFB) for voice calls.

15. An engineer is analyzing the performance of OFDMA compared to CDMA for a 4G LTE system. The engineer notes that both schemes allow multiple users to access the network simultaneously. However, OFDMA, by using a large number of orthogonal subcarriers, is inherently more robust against a specific type of interference that severely degrades CDMA performance, especially in high-data-rate scenarios. This interference arises from signals taking multiple paths to the receiver. What is this phenomenon, and how does OFDMA's architecture inherently provide resilience against it?

a) Near-far effect; mitigated by using a higher chip rate.
b) Co-channel interference; mitigated by frequency hopping.
c) Adjacent channel interference; mitigated by using guard bands between subcarriers.
d) Multipath fading; mitigated by using a cyclic prefix and long symbol duration.

16. A student is preparing a presentation on the evolution of cellular networks, specifically comparing the key performance indicators of 1G through 5G. One slide focuses on the peak data rates. The student correctly notes that 1G had no data capability, 2G (GSM) introduced data at low speeds (9.6 kbps), 3G (UMTS) offered up to 2 Mbps, and 4G (LTE) offered hundreds of Mbps. However, the student is confused about the enhancements made to 3G to achieve higher data rates before 4G was introduced. Which of the following technologies was introduced in 3G (UMTS) to significantly increase downlink data rates, forming a bridge towards 4G?

a) Global System for Mobile communications (GSM)
b) High-Speed Downlink Packet Access (HSDPA)
c) High-Speed Circuit Switched Data (HSCSD)
d) General Packet Radio Service (GPRS)

17. A mobile device is communicating with a base station. In the context of radio interface protocols, the operation of the system is often described in terms of a layered architecture. The physical layer (Layer 1) is responsible for the actual transmission and reception of bits over the air. The data link layer (Layer 2) is responsible for providing a reliable link between the mobile station and the base station. One crucial function of the data link layer is the segmentation and reassembly of data, as well as error control. In GSM, which sub-layer of Layer 2 is specifically responsible for managing the logical link between the MS and the BSS, performing functions like error correction and flow control?

a) The Medium Access Control (MAC) layer
b) The Mobility Management (MM) layer
c) The Radio Link Control (RLC) layer
d) The Physical Layer (L1)

18. A wireless communication researcher is studying the localization techniques used in cellular networks. In GSM, location services can be based on different principles. Cell Identity (CI) is the simplest method, where the mobile's location is known to be within the coverage area of a particular cell. However, more accurate methods like Time of Arrival (TOA) and Time Difference of Arrival (TDOA) are also used. The researcher is analyzing a scenario where a mobile device transmits a signal, and three base stations measure the signal's arrival time. TDOA calculates the difference in arrival times at pairs of base stations. What is the fundamental principle used to determine the location in TDOA-based localization?

a) The mobile's location is determined by the angle of arrival (AOA) of the signal at each base station.
b) The mobile's location is the centroid of the three base stations.
c) The mobile's location is the intersection point of hyperbolas, which represent the difference in distance from the mobile to two base stations.
d) The mobile's location is the intersection point of three circles, each centered at a base station.

19. A network operator is planning a conference in a large convention center. To handle the anticipated high volume of mobile data traffic, they decide to install a small-cell system. They are debating between using a femtocell and a picocell. Both are types of small cells, but they serve different purposes. The convention center is a publicly accessible space with an expected traffic density of hundreds of users. Which of the following correctly describes the primary difference in the deployment scenario and capacity of a femtocell versus a picocell in this context?

a) A picocell is a public small cell covering a larger area (e.g., a convention center floor) and supporting more users than a residential femtocell.
b) Femtocells use licensed spectrum, whereas picocells use unlicensed spectrum like Wi-Fi.
c) Both femtocells and picocells have the same range and capacity; the difference is only in their backhaul connection.
d) A femtocell is typically used for large public spaces, while a picocell is designed for residential or small office use.

20. A systems engineer is analyzing the handover process in a GSM network, focusing on the role of the Mobile Station (MS). The MS continuously measures the signal strength and quality of the serving cell and its neighboring cells. These measurements are reported to the network. While the network largely controls the handover decision, the MS can initiate a specific type of handover. In GSM, what is this type of handover, and under what condition does it typically occur?

a) Handover initiated by the MS; this is a standard procedure known as "Mobile Assisted Handover" (MAHO).
b) Asynchronous handover; initiated to balance traffic load between cells.
c) Synchronous handover; initiated when the MS detects a stronger neighboring cell.
d) Emergency handover; initiated by the MS when the serving cell's signal quality drops below a critical threshold.

21. A university professor is explaining the concept of multiple access techniques to a class. The professor uses a metaphor: FDMA is like having separate radio stations on different frequencies; TDMA is like people taking turns talking in a conversation; CDMA is like a cocktail party where everyone speaks a different language and the listener can tune in to one specific language. Based on this metaphor, the professor asks the class to identify the key challenge of the CDMA system (the cocktail party). Which of the following best describes the "near-far" problem in CDMA in this context?

a) The challenge of understanding a language if too many people are speaking the same language.
b) The need to schedule a time for each person to speak.
c) The difficulty in hearing a quiet speaker who is far away when a loud speaker is very close to you, even though they are speaking different languages.
d) The problem of interference from overlapping frequencies in the radio spectrum.

22. An intern at a telecom company is trying to understand the circuit-switched and packet-switched cores in a GSM/GPRS network. The intern is confused about the path a data packet takes when a user is browsing the web. The user sends a request which goes from the mobile station to the base station and then to the BSC. The intern correctly identifies that for GPRS, the data then goes to the SGSN. What is the final network element that the packet traverses to exit the mobile operator's core network and reach the public internet?

a) The Base Station Controller (BSC)
b) The Mobile Switching Center (MSC)
c) The Gateway GPRS Support Node (GGSN)
d) The Home Location Register (HLR)

23. A wireless network analyst is comparing the characteristics of Code Division Multiple Access (CDMA) and Orthogonal Frequency Division Multiple Access (OFDMA). They note that while CDMA uses a spreading code to spread the signal over a wide bandwidth, OFDMA uses multiple orthogonal subcarriers. The analyst is evaluating the spectral efficiency of both in a multi-path environment. A major advantage of OFDMA over CDMA in this context is its resilience to frequency-selective fading. What is the fundamental reason for this increased resilience?

a) OFDMA divides the wideband channel into many narrowband subcarriers, each experiencing flat fading, making equalization simpler.
b) OFDMA is less susceptible to the near-far problem.
c) OFDMA does not require any power control.
d) OFDMA uses a higher chip rate than CDMA.

24. A network engineer is designing the security architecture for a new mobile network. The engineer wants to ensure mutual authentication between the mobile device and the network. They are studying the GSM and UMTS authentication mechanisms. In GSM, the network authenticates the user using a challenge-response mechanism based on the Ki key on the SIM. However, UMTS introduced a more robust mechanism to also authenticate the network. What is the key difference that allows the UMTS mobile station (USIM) to validate the network's authenticity?

a) UMTS requires the user to enter a PIN for every authentication event.
b) UMTS uses a new ciphering algorithm (A5/1).
c) UMTS uses a longer encryption key (Kc).
d) UMTS includes an Authentication Token (AUTN) in the challenge, allowing the MS to verify the network's identity.

25. A telecommunications operator is planning a new network deployment. They are considering using High-Speed Circuit Switched Data (HSCSD) to offer data services. HSCSD is an enhancement of GSM that allows a single mobile station to use multiple time slots simultaneously. An engineer argues that HSCSD is inefficient compared to GPRS. What is the core architectural disadvantage of HSCSD that leads to this inefficiency for typical data traffic?

a) HSCSD has a higher latency than GPRS.
b) HSCSD, being a circuit-switched technology, reserves resources (time slots) for the duration of the connection, regardless of whether data is being transmitted.
c) HSCSD is not compatible with existing GSM infrastructure.
d) HSCSD uses a less efficient modulation scheme than GPRS.

26. A radio planner is calculating the capacity of an FDMA system. The system has a total bandwidth of 12.5 MHz. Each channel requires 30 kHz of bandwidth. The system uses a 4-cell reuse pattern (N=4). What is the total number of channels available per cell if the operator wants to maximize capacity while adhering to the reuse pattern?

a) 104.17 (approx. 104) channels
b) 416.67 (approx. 417) channels
c) 1666.67 (approx. 1667) channels
d) 30 channels

27. A mobile subscriber is driving on a highway and is using a GSM phone for a voice call. The call is handed over from one cell to another. The handover process involves several signaling messages between the mobile station (MS) and the network. Which of the following best describes the primary purpose of the "Handover Command" message sent from the network to the MS?

a) To instruct the MS on which new frequency and time slot to use for the connection.
b) To request the MS to measure the signal quality of the new cell.
c) To confirm that the MS has successfully disconnected from the old cell.
d) To inform the MS that the network has dropped the call due to an error.

28. An engineer is comparing the power control mechanisms of CDMA and GSM. In GSM, power control is relatively slow and is mainly used to reduce interference and save battery power. In CDMA, power control is fast and stringent. Why is fast and accurate power control so critical for the proper operation of a CDMA system?

a) To prevent the mobile station's battery from draining too quickly.
b) To ensure that each mobile station's signal arrives at the base station with the same power level, thereby maximizing the system capacity.
c) To reduce the frequency of handovers between cells.
d) To allow the system to use a lower chip rate.

29. A researcher is evaluating the performance of OFDMA in an LTE system. The researcher notes that OFDMA uses a Cyclic Prefix (CP) to mitigate Inter-Symbol Interference (ISI). The CP is a copy of the end of the symbol appended to the beginning. However, the CP consumes a portion of the available bandwidth and power. An engineer argues that for a high-data-rate system in a highly dispersive (large multipath delay spread) environment, a longer CP is necessary. What is the primary trade-off in choosing the length of the Cyclic Prefix?

a) A longer CP increases data rate but reduces coverage.
b) A longer CP reduces Peak-to-Average Power Ratio (PAPR) but increases latency.
c) A longer CP mitigates ISI more effectively but reduces spectral efficiency.
d) A longer CP simplifies the receiver design but increases the battery drain on the mobile device.

30. A network operator is looking to deploy a technology that can offer high-speed data services using the existing GSM infrastructure. They want a solution that is packet-switched and efficiently uses the radio resources for bursty data traffic. The operator's main goal is to allow multiple users to share the same time slot and to provide a data rate that is significantly higher than GSM's 9.6 kbps. Which technology, which was a key step towards 3G, fulfills these requirements?

a) Universal Mobile Telecommunications System (UMTS)
b) Enhanced Data rates for GSM Evolution (EDGE)
c) General Packet Radio Service (GPRS)
d) High-Speed Circuit Switched Data (HSCSD)

31. A student of wireless networks is listing the key services offered by GSM. They include teleservices (e.g., voice), bearer services (e.g., data transmission), and supplementary services (e.g., call forwarding). Which of the following is a correct classification of a GSM service?

a) Call waiting is a teleservice.
b) Voice telephony is a teleservice.
c) Data transmission (circuit-switched) is a supplementary service.
d) Short Message Service (SMS) is a bearer service.

32. An engineer is studying the architecture of the UMTS network. Unlike GSM's circuit-switched core, UMTS was designed to support both circuit-switched and packet-switched services from the start. The UMTS core network is divided into two domains. What are these two domains, and what is their primary purpose?

a) The Radio Access Network (RAN) and the Core Network (CN).
b) The Base Station Subsystem (BSS) and the Network Switching Subsystem (NSS).
c) The User Equipment (UE) domain and the Network domain.
d) The Circuit Switched (CS) domain for voice and the Packet Switched (PS) domain for data.

33. A technology analyst is reviewing the characteristics of the Digital Enhanced Cordless Telecommunications (DECT) standard. DECT is often used for wireless office systems and home cordless phones. The analyst is explaining to a colleague how DECT differs from cellular standards like GSM. Which of the following is a key feature of DECT that makes it suitable for its intended short-range, high-density environment?

a) Its use of a large macro-cell layout.
b) Its support for multiple simultaneous calls per base station and its efficient TDMA structure.
c) Its ability to handover at high vehicular speeds.
d) Its reliance on a single, global licensed spectrum band.

34. A mobile device is located in a city and is trying to establish a connection. In a GSM network, the mobile station goes through several steps, including searching for a carrier, synchronizing, and registering with the network. The process involves the broadcast of certain information by the Base Transceiver Station (BTS). Which of the following is broadcast on the Broadcast Control Channel (BCCH) and is essential for the mobile station to initially identify the network?

a) The International Mobile Equipment Identity (IMEI)
b) The Ciphering Key (Kc)
c) The Location Area Identity (LAI) and the Cell Identity (CI)
d) The Temporary Mobile Subscriber Identity (TMSI)

35. A network administrator is analyzing a scenario where a subscriber's GSM phone is switched off or moves out of coverage. The network performs an implicit detach, marking the subscriber as inactive. When the subscriber turns the phone back on, it performs a Location Update to inform the network. However, the network also keeps track of the subscriber's state using a timer. What is the name of this timer, and what function does it serve in the VLR?

a) T3122 timer; it controls the frequency of paging attempts.
b) T3122 timer; it controls the frequency of cell reselection.
c) T3212 timer; it controls the frequency of handover attempts.
d) T3212 timer; it controls the frequency of periodic location updates.

36. A wireless communication professor is explaining the evolution of mobile networks from an architectural perspective. The professor notes that 2G GSM used a circuit-switched architecture with a dedicated MSC, while 3G UMTS introduced a packet-switched core with SGSN and GGSN. The professor then describes how 4G LTE evolved to a flat, all-IP architecture. What was a key architectural change in LTE that simplified the network and reduced latency?

a) The replacement of the BSC and RNC functions into the eNodeB, removing the separate radio network controller layer.
b) The combining of the SGSN and GGSN functions into a single Serving Gateway (S-GW) and Packet Data Network Gateway (P-GW).
c) The elimination of the Base Station (eNodeB) and the use of direct satellite communication.
d) The introduction of a new, more powerful MSC (MSC Server).

37. An engineer is tasked with designing a system that can serve a very large number of users in a small area, like a stadium. The engineer decides to use an SDMA-based approach. Which of the following is a practical and significant challenge in the implementation of SDMA that the engineer must address?

a) The requirement for a large, continuous block of spectrum.
b) The need for complex antenna arrays and sophisticated digital signal processing for beamforming.
c) The complexity of time-slot synchronization.
d) The inherent susceptibility to the near-far problem.

38. A network operator is evaluating the security of its legacy GSM network. A known vulnerability of GSM is that the encryption between the mobile station and the base station is only in one direction (from MS to BTS) and can be compromised. Which of the following is NOT a vulnerability or limitation of the GSM security architecture?

a) The vulnerability of the A5/1 encryption algorithm.
b) The lack of mutual authentication (the network is not authenticated by the MS).
c) The use of a relatively short key length (Kc) for encryption.
d) The use of an end-to-end encryption mechanism between two mobile stations.

39. An engineer is analyzing the different multiple access techniques used in cellular systems. They are preparing a presentation that highlights the advantages and disadvantages of each. Which of the following correctly describes a major disadvantage of TDMA that is not a primary concern for FDMA?

a) The susceptibility to co-channel interference.
b) The requirement for complex filtering.
c) The need for guard bands between channels.
d) The need for precise synchronization between the mobile station and the base station.

40. A radio network planning team is choosing between FDD and TDD for a new network deployment. The team is analyzing the spectrum requirements and the type of services to be offered. They have a limited amount of paired spectrum available (symmetric bands) but a large amount of unpaired spectrum. Which duplexing scheme would be more suitable for this situation, and why?

a) FDD, because it provides better spectral efficiency than TDD.
b) FDD, because it requires only a single frequency band.
c) TDD, because it can operate in unpaired spectrum and does not require symmetric frequency bands for uplink and downlink.
d) TDD, because it is less susceptible to interference than FDD.

41. A student is comparing the GSM, UMTS, and LTE radio interfaces. They note that GSM uses TDMA and FDMA, UMTS uses CDMA, and LTE uses OFDMA. The student observes that with each generation, the physical layer has become more complex and efficient. What was the primary motivation for the shift from CDMA (in UMTS) to OFDMA (in LTE)?

a) The requirement to reduce the number of base stations.
b) The desire to use a narrower channel bandwidth.
c) The need for a simpler receiver design in the mobile device.
d) The ability to support higher data rates and provide better resilience to frequency-selective fading.

42. A network architect is explaining the concept of "localization" in GSM. The architect describes three levels of location granularity: Cell Global Identity (CGI), Location Area Identity (LAI), and the Mobile Station's position. The architect emphasizes that paging is performed at the level of a Location Area. What is the primary advantage of dividing the network into Location Areas for paging?

a) It improves the voice quality of the calls.
b) It reduces the number of handovers.
c) It allows the network to allocate dedicated frequencies to each user.
d) It reduces the amount of signaling required to find a mobile station, as the network only pages in the LAI rather than the entire network.

43. An engineer is evaluating the performance of GPRS. They note that GPRS introduces a new channel structure on the air interface, using packet data channels (PDCHs). These PDCHs are allocated dynamically based on traffic demand. What is the key advantage of this dynamic allocation compared to the dedicated channels used in GSM?

a) It guarantees a fixed data rate for each user.
b) It eliminates the need for the SGSN.
c) It allows for a more efficient use of radio resources, especially for bursty data traffic.
d) It reduces the complexity of the mobile device.

44. A technician is setting up a new DECT system for a small office. They are aware that DECT uses a TDMA scheme with 12 time slots per frame. For a cordless phone, one slot is used for the downlink from the base station to the handset, and another slot is used for the uplink. This makes DECT a true TDD system. What is the frame length of a DECT frame, which determines the latency of the system?

a) 10 ms
b) 20 ms
c) 1 ms
d) 4.615 ms

45. A business analyst is preparing a report on the market impact of different generations of mobile networks. They note that the introduction of 3G (UMTS) was a significant milestone. What was the primary service or capability that UMTS brought to the mass market, distinguishing it from 2G (GSM/GPRS)?

a) The use of a circuit-switched core network.
b) The introduction of the Short Message Service (SMS).
c) The ability to support mobile broadband data services with significantly higher speeds.
d) The introduction of digital encryption for voice calls.

46. A network security expert is analyzing the vulnerabilities in the GSM network. They point out that a major weakness is the lack of network authentication. What type of attack does this vulnerability specifically enable, where a malicious actor impersonates a legitimate base station?

a) A Man-in-the-Middle (MITM) attack
b) A Replay attack
c) A Denial of Service (DoS) attack
d) An Eavesdropping attack

47. A mobile network operator is planning to deploy a 5G network. A key feature of 5G is its support for a wide range of use cases, from enhanced Mobile Broadband (eMBB) to Ultra-Reliable Low-Latency Communications (URLLC). The operator is considering how to partition its network resources to guarantee specific performance levels for each use case. Which 5G core network concept allows the operator to create, isolated, end-to-end logical networks on the same physical infrastructure to meet these diverse requirements?

a) Network Slicing
b) Software-Defined Networking (SDN)
c) Massive MIMO
d) Network Function Virtualization (NFV)

48. A radio engineer is measuring the signal parameters of an OFDMA signal. The engineer notes that each subcarrier in an OFDMA symbol carries a portion of the data. The data is transmitted in parallel, which results in a longer symbol duration compared to a single-carrier system with the same data rate. What is the primary benefit of this longer symbol duration?

a) It makes delay better.
b) It reduces the Peak-to-Average Power Ratio (PAPR).
c) None of the above
d) Both A and B

49. A network operator is comparing the resource allocation mechanisms of GSM and GPRS. In GSM, a time slot is dedicated to a single user for the duration of the call. In GPRS, multiple users can share the same time slot, and resources are allocated on a per-packet basis. An engineer argues that this makes GPRS significantly more efficient for data traffic. However, a manager counters that this efficiency comes at the cost of increased complexity. Which of the following is a direct consequence of GPRS's shared resource model that increases the complexity of the system compared to GSM?

a) The need for a separate physical channel for each user's data.
b) The requirement for a more complex scheduling algorithm to manage

50. A wireless communications student is analyzing the cellular concept and frequency reuse. The student understands that in FDMA, the total available bandwidth is divided into channels, and these channels are reused in different cells at a distance far enough to keep co-channel interference (CCI) at an acceptable level. The student is studying a system with a reuse factor of 1/3 (N=3). Which of the following correctly describes the relationship between the frequency reuse factor and the system capacity in this specific scenario?

a) A reuse factor of 1/3 means the system uses 1/3 of the bandwidth for control channels and 2/3 for traffic channels.

b) A reuse factor of 1/3 means each cell uses 3 times the total channels, providing the highest capacity.

c) A reuse factor of 1/3 means each cell uses 1/3 of the total channels, and the cluster size is 3, which maximizes capacity but increases CCI.

d) A reuse factor of 1/3 implies that all cells use the same frequency set, resulting in no CCI.

51. A network designer is studying the handover process in GSM. The designer notes that there are different types of handovers: synchronous and asynchronous. In a synchronous handover, the mobile station (MS) already knows the timing advance required for the new cell, making the handover faster. In an asynchronous handover, the MS must acquire the timing of the new cell before transmitting. The designer is analyzing a scenario where an MS is moving from a cell managed by BSC-1 to a cell managed by BSC-2. Both BSCs are synchronized to the same clock source. What type of handover will occur, and what is the primary benefit of this synchronization?

a) Hard handover; it is always asynchronous.

b) Synchronous handover; it speeds up the handover process and reduces the disruption time.

c) Inter-MSC handover; it ensures seamless roaming between different operators.

d) Asynchronous handover; it allows the MS to measure the new cell before the handover.

52. A telecommunications historian is analyzing the transition from 3G (UMTS) to 4G (LTE). The historian notes that one of the key drivers for LTE was the need for a more efficient core network architecture to handle the exponential growth of mobile data traffic. In UMTS, the core network had separate Circuit-Switched (CS) and Packet-Switched (PS) domains. LTE adopted a flat, all-IP architecture, eliminating the CS domain. What was the primary consequence of this architectural simplification in LTE?

a) It made voice calls more reliable.

b) It required all voice calls to be routed through the PS domain using VoLTE, which increased network complexity.

c) It eliminated the need for handovers between cells.

d) It reduced the latency and signaling overhead, improving the user experience for data services.

53. A radio frequency engineer is evaluating the performance of a CDMA system. The engineer observes that the system is experiencing severe interference from a neighboring cell operating on the same frequency. This is known as inter-cell interference, which is a major limitation of CDMA systems. The engineer is considering implementing a soft handover to mitigate this issue. In a soft handover, the mobile station is simultaneously connected to two or more base stations. How does soft handover help reduce interference in a CDMA system?

a) By transmitting at a higher power to overcome interference.

b) By combining signals from multiple base stations, the mobile station can use macro-diversity to improve signal quality and reduce the required transmit power.

c) By disconnecting from the interfering base station before connecting to the new one.

d) By switching the mobile station to a different frequency band.

54. A network planner is tasked with deploying a network in a rural area with low user density but large coverage requirements. The planner is comparing GSM and UMTS for this scenario. GSM uses TDMA/FDMA with relatively high transmit power per channel, while UMTS uses CDMA. The planner notes that GSM cells can cover a larger area than UMTS cells because of a specific characteristic of the air interface. What is this characteristic that gives GSM an advantage in coverage?

a) GSM does not require power control, unlike UMTS.

b) GSM uses a more robust modulation scheme than UMTS.

c) GSM uses a higher frequency band than UMTS, leading to better propagation.

d) GSM's TDMA/FDMA scheme allows for a higher peak transmit power per time slot, providing better link budget.

55. A researcher is comparing the data rates supported by HSCSD and GPRS. HSCSD can aggregate multiple time slots, offering up to 57.6 kbps in GSM. GPRS can also aggregate multiple time slots and uses a more efficient coding scheme, offering up to 171.2 kbps. However, a key difference is that HSCSD allocates resources continuously, while GPRS uses a packet-switched approach. A student asks why GPRS is considered more efficient than HSCSD despite HSCSD being able to provide a similar maximum data rate. What is the correct justification for this?

a) GPRS is more efficient because it uses a dedicated channel for each user.
b) HSCSD is more efficient because it has a lower latency.
c) GPRS is more efficient because it allocates radio resources only when there is data to send, leading to better spectral utilization.
d) GPRS is more efficient because it requires less complex hardware in the mobile station.

56. A mobile device is performing a Location Update in a GSM network. The process involves the Mobile Station (MS), the Base Station System (BSS), the Visitor Location Register (VLR), and the Home Location Register (HLR). The MS sends a Location Update Request to the network. The network identifies that the MS has moved to a new Location Area (LA) and needs to update its location information. Which sequence correctly describes the interaction between the VLR and HLR during this procedure?

a) The MS directly contacts the HLR to update its location, bypassing the VLR.
b) The BSC updates the HLR, which then informs the VLR.
c) The VLR directly updates the HLR with the new LAI, and the HLR updates the VLR with subscriber data.
d) The VLR sends the location update to the BSS, which then forwards it to the HLR.

57. A systems engineer is evaluating the security mechanisms of GSM, UMTS, and LTE. The engineer notes that GSM uses a one-way authentication mechanism, UMTS introduced mutual authentication, and LTE further enhanced security by using a more robust key hierarchy and encryption algorithms. A core vulnerability that still exists in all these systems is that the encryption is only between the mobile device and the base station, not end-to-end. Which of the following is a direct implication of this limitation?

a) Voice calls cannot be encrypted.
b) Data is protected only on the air interface; it is decrypted at the base station and then travels unprotected through the core network.
c) The network operator cannot decrypt user data for lawful interception.
d) The mobile device cannot authenticate the network.

58. A radio network engineer is analyzing the power control algorithm in a WCDMA (UMTS) system. The system uses fast closed-loop power control to combat fast fading. The mobile station measures the signal-to-interference ratio (SIR) of the received downlink signal and sends Transmit Power Control (TPC) commands to the base station every 0.667 ms. What is the primary reason for the high frequency of these TPC commands in a UMTS system?

a) To reduce the battery consumption of the mobile station.
b) To compensate for the rapid fluctuations in signal strength caused by multipath propagation (fast fading).
c) To compensate for the slow variations in path loss due to shadowing.
d) To coordinate handovers between cells.

59. A student is preparing a report on the evolution of mobile network architectures. The student notes that GSM had a centralized architecture with a Mobile Switching Center (MSC) managing mobility and switching. UMTS introduced a separate Radio Network Controller (RNC) between the base station (Node B) and the core network. LTE simplified this by eliminating the RNC and distributing its functions to the eNodeB. What was the primary motivation for this architectural simplification in LTE?

a) To increase the coverage area of each cell.
b) To reduce the cost of base stations.
c) To reduce network latency and improve scalability by creating a flatter network architecture.
d) To simplify the handover process.

60. A telecommunications consultant is advising a client on the adoption of 5G technology. The client is interested in using 5G for industrial automation, which requires extremely low latency and high reliability (URLLC). The consultant explains that 5G achieves this through several key features. Which of the following 5G features is specifically designed to minimize the time it takes for a packet to travel from the device to the network and back?

a) Massive MIMO to increase spectral efficiency through spatial multiplexing.
b) Network slicing to create virtualized, service-specific logical networks.
c) Enhanced Mobile Broadband (eMBB) to provide higher peak data rates.
d) Multi-access Edge Computing (MEC) that places compute resources close to the user to shorten round-trip paths.

61. An engineer is analyzing the physical layer of an LTE system. The engineer understands that LTE uses OFDMA in the downlink and SC-FDMA (Single Carrier Frequency Division Multiple Access) in the uplink. The engineer is asked why SC-FDMA is preferred for the uplink instead of OFDMA. What is the primary technical reason for this choice?

a) SC-FDMA provides better coverage than OFDMA.
b) SC-FDMA is less susceptible to multipath fading than OFDMA.
c) SC-FDMA is simpler to implement than OFDMA.
d) SC-FDMA has a lower Peak-to-Average Power Ratio (PAPR) than OFDMA, which allows the mobile device's power amplifier to operate more efficiently.

62. A network analyst is studying the performance of different multiple access schemes. The analyst notes that CDMA is a "soft capacity" system, while FDMA and TDMA are "hard capacity" systems. What does this mean in terms of system planning and operation?

a) FDMA and TDMA have no capacity limits.
b) In CDMA, the capacity is fixed and determined by the number of available codes.
c) In CDMA, adding a new user increases the interference to all other users, reducing the quality; there is no fixed limit on the number of users, only a degradation in performance.
d) CDMA capacity is independent of interference.

63. A mobile subscriber is using a GSM phone in a train traveling at 250 km/h. The subscriber is experiencing frequent call drops. A network engineer is investigating the problem and suspects that the handover process is not fast enough. In GSM, there are several parameters that control handover, including the "Power Budget" and the "Signal Strength" thresholds. Which of the following GSM handover algorithms is most likely failing, and what is its primary function in this high-mobility scenario?

a) The power budget handover; it triggers a handover when a neighboring cell provides a better signal, which may be happening too slowly.
b) The distance-based handover; it is using the Timing Advance to calculate distance.
c) The traffic handover; it is not triggered because there is no congestion.
d) The interference-based handover; it is not triggered because the interference is low.

64. A network designer is comparing the core network elements of GSM and UMTS. In GSM, the MSC performs both call switching and mobility management. In UMTS, the MSC is split into the MSC Server (MSC-S) and the Media Gateway (MGW). What is the primary benefit of this separation in UMTS?

a) It allows for a more efficient handling of circuit-switched and packet-switched traffic simultaneously.
b) It eliminates the need for the HLR.
c) It separates the control plane (signaling) from the user plane (bearer traffic), enabling a more flexible and scalable network architecture.
d) It simplifies the architecture of the Radio Access Network.

65. A radio propagation expert is analyzing the impact of the environment on signal quality in an OFDMA system. The expert is studying the effects of multipath propagation, where the signal arrives at the receiver via multiple paths with different delays. The expert notes that the OFDMA receiver uses a Fast Fourier Transform (FFT) to demodulate the subcarriers. What condition must be satisfied to ensure orthogonality between the subcarriers and to avoid Inter-Carrier Interference (ICI)?

a) The subcarriers must be spaced at multiples of the coherence bandwidth.

b) The symbol duration must be much larger than the maximum delay spread, and the cyclic prefix must be longer than the delay spread.

c) The transmit power must be high.

d) The number of subcarriers must be a power of two.

66. A business analyst is evaluating the cost-effectiveness of deploying a DECT system versus a Wi-Fi network for a large office building. The building has a high density of users, all of whom need to make voice calls and access the internet. DECT is designed for voice with guaranteed quality of service, while Wi-Fi is designed for data. The analyst notes that DECT operates in a licensed spectrum, while Wi-Fi uses unlicensed spectrum. What is a key advantage of DECT over Wi-Fi in this enterprise environment?

a) DECT does not require a base station.

b) DECT can handle a higher density of simultaneous voice calls with guaranteed quality, as it is optimized for voice and uses a dedicated TDMA structure.

c) DECT provides higher data rates than Wi-Fi.

d) DECT provides better security due to the licensed spectrum.

67. A telecommunications regulator is studying the spectrum allocation for different mobile network generations. The regulator notes that 2G GSM operated primarily in the 900 MHz and 1800 MHz bands. 3G UMTS operated in the 2100 MHz band. 4G LTE operates in a wide range of frequencies, including 700 MHz, 1800 MHz, and 2600 MHz. 5G introduces millimeter-wave (mmWave) bands above 24 GHz. What is the primary trade-off when using higher frequency bands like mmWave for 5G?

a) Higher frequency bands require less complex antennas.

b) Higher frequency bands provide better coverage but lower data rates.

c) Higher frequency bands are less susceptible to interference.

d) Higher frequency bands offer greater available bandwidth and higher data rates but suffer from higher path loss and poorer penetration.

68. A network engineer is configuring the authentication parameters in a GSM network. The engineer retrieves the Authentication Center (AuC) database. The database contains the International Mobile Subscriber Identity (IMSI) and the subscriber's secret key (Ki). The engineer initiates an authentication request. Which of the following correctly describes the sequence of events in the GSM authentication process?

a) The MSC sends a random number (RAND) to the MS, which uses Ki to compute a response (SRES) and sends it back for verification.

b) The HLR sends the encryption key (Kc) directly to the MS.

c) The MS authenticates the network by checking the LAI.

d) The MS sends its Ki to the network for verification.

69. A student is researching the technical specifications of different wireless standards. The student is comparing GSM (TDMA/FDMA), IS-95 (CDMA), and LTE (OFDMA). The student notes that each uses a different duplexing scheme. GSM uses FDD, IS-95 uses FDD, and LTE can use both FDD and TDD. What is a significant advantage of TDD over FDD in LTE for asymmetric data traffic?

a) TDD provides better coverage than FDD.

b) TDD does not require synchronization between base stations.

c) TDD allows the operator to use a single frequency band and dynamically adjust the allocation of time slots between uplink and downlink.

d) TDD uses less complex RF filters.

70. A mobile network operator is experiencing congestion in its GSM network's control channels. The Signalling Channel (SCH) and Broadcast Control Channel (BCCH) are broadcast channels, while the Stand-alone Dedicated Control Channel (SDCCH) is a point-to-point control channel. A large number of mobile stations are performing Location Updates and call setup attempts, causing SDCCH congestion. Which of the following is a valid strategy to alleviate this specific congestion?

a) Increasing the number of time slots allocated for SDCCH in the cell.
b) Disabling encryption on the traffic channels.
c) Reducing the transmit power of the BTS.
d) Increasing the frequency reuse factor.

71. A network researcher is evaluating the localization accuracy of different techniques in a cellular network. The researcher compares Cell Identity (CI), Time of Arrival (TOA), Time Difference of Arrival (TDOA), and Angle of Arrival (AOA). Which technique provides the most accurate location estimation in an urban environment with good multipath propagation, and what is its primary limitation?

a) TDOA, but it is limited by the need for at least three base stations and is susceptible to multipath errors.
b) TOA, but it is limited by the need for precise synchronization between the mobile and the base station.
c) CI, but it is very inaccurate.
d) AOA, but it requires complex antenna arrays at the base station.

72. An engineer is analyzing the handover performance in a GSM network. The engineer observes that when a mobile station crosses a cell boundary, there is a brief interruption in the call. This is known as a hard handover (break-before-make). In a CDMA (UMTS) system, the call does not drop during a handover because it uses a soft handover (make-before-break). The engineer is asked to explain why GSM uses hard handover instead of soft handover. What is the fundamental reason for this difference?

a) GSM does not implement power control, so soft handover would be ineffective.
b) GSM uses narrowband TDMA/FDMA with different carrier frequencies in adjacent cells, and the mobile cannot maintain parallel links on the same channel, so handover must be break-before-make.
c) GSM uses spread-spectrum codes, which naturally support make-before-break without frequency planning.
d) GSM base stations are tightly synchronized, eliminating the need for make-before-break.

73. A telecommunications manager is presenting a proposal to upgrade the company's network from 3G to 4G LTE. A board member asks what the key benefits of LTE are over UMTS in terms of the user experience. Which of the following is a correct and fundamental advantage of LTE over UMTS?

a) LTE eliminates the need for base stations.
b) LTE has a flatter architecture with significantly lower latency and higher peak data rates.
c) LTE uses the same CDMA air interface as UMTS but with more codes.
d) LTE uses circuit switching for voice, making it more reliable.

74. A radio network planner is designing a cellular network and must choose between a 4-cell reuse pattern (N=4) and a 7-cell reuse pattern (N=7) for an FDMA system. The planner wants to maximize the number of users per cell and must also ensure that the co-channel interference remains within acceptable limits. Which statement correctly describes the trade-off between these two reuse patterns?

a) N=4 provides better signal quality than N=7.
b) N=4 provides higher capacity (more users per cell) but suffers from higher co-channel interference than N=7.
c) The capacity and co-channel interference are the same for both patterns.
d) N=7 provides higher capacity than N=4.

75. A student is learning about the radio interface protocols in GSM. The student understands that the physical layer (L1) is responsible for the transmission of bits over the air, and the data link layer (L2) provides reliable data transfer. The student is studying the specific channels used on the GSM air interface. The Broadcast Control Channel (BCCH) is a unidirectional channel from the BTS to the MS, while the Dedicated Control Channel (DCCH) is bidirectional. Which layer is responsible for managing these logical channels and mapping them to the physical channels (time slots)?

a) Medium Access Control (MAC) sublayer
b) Physical Layer (L1)
c) Radio Resource Management (RR) sublayer
d) Mobility Management (MM) sublayer

76. A network operator is planning to deploy a 5G network in a dense urban area and is considering using a massive MIMO (Multiple-Input Multiple-Output) antenna system. Massive MIMO uses a large number of antenna elements at the base station to serve multiple users simultaneously on the same time-frequency resources. What is the primary benefit of massive MIMO in this scenario?

a) It reduces the cost of the base station.
b) It allows the base station to focus energy towards specific users, increasing spectral efficiency and reducing interference.
c) It simplifies the handover process.
d) It requires less spectrum than traditional MIMO.

77. A researcher is analyzing the evolution of mobile network security. The researcher notes that GSM used the A5/1 cipher for encryption, which was later found to be weak. UMTS used the A5/3 cipher (Kasumi), which is more robust. LTE uses the AES (Advanced Encryption Standard) algorithm for encryption. What is a key characteristic of AES that makes it superior to A5/1 for securing mobile communications?

a) AES is a block cipher with a longer key length, making it more resistant to brute-force attacks.
b) AES is designed for circuit-switched networks only.
c) AES is a stream cipher, making it faster than A5/1.
d) AES uses a fixed 64-bit key.

78. A technical analyst is comparing the capabilities of GPRS and EDGE. Both are enhancements to GSM, but EDGE offers higher data rates by introducing a new modulation scheme called 8-PSK (Phase Shift Keying) in addition to the GSM standard GMSK (Gaussian Minimum Shift Keying). What is the trade-off involved in using 8-PSK instead of GMSK?

a) 8-PSK provides better coverage than GMSK.
b) 8-PSK is less complex to implement than GMSK.
c) 8-PSK uses less bandwidth than GMSK.
d) 8-PSK transmits 3 bits per symbol (higher data rate) but is less robust to noise and interference, requiring a better signal-to-noise ratio.

79. A mobile network engineer is troubleshooting a situation where a mobile station is not able to make a call. The engineer checks the signaling and finds that the mobile station is not receiving the TMSI (Temporary Mobile Subscriber Identity) from the network during the location update. The TMSI is used instead of the IMSI to identify the subscriber on the air interface for privacy. What is the direct consequence of not assigning a TMSI to a mobile station?

a) The mobile station will not be able to authenticate.
b) The mobile station will not be able to roam.
c) The mobile station will not be able to perform a handover.
d) The mobile station's IMSI must be transmitted over the air in clear text, compromising the subscriber's privacy.

80. A professor is explaining the concept of spectral efficiency to a class of telecommunications students and asks the students to compare the spectral efficiency of FDMA, TDMA, and CDMA. Which of the following is a generally true statement about the potential spectral efficiency of these access methods?

a) TDMA has higher spectral efficiency than FDMA because it does not require guard bands.
b) CDMA has the potential for higher spectral efficiency than FDMA and TDMA because it reuses the entire frequency band in every cell (reuse factor of 1), but this is limited by interference.
c) FDMA has the highest spectral efficiency because it uses a dedicated frequency for each user.
d) All three access methods have the same spectral efficiency.

81. A network designer is evaluating the performance of HSCSD (High-Speed Circuit Switched Data). A manager argues that HSCSD is not suitable for modern data applications. Which of the following is the most significant architectural limitation of HSCSD that makes it unsuitable for bursty, intermittent data traffic?

a) It reserves multiple time slots exclusively for a single user for the entire duration of the connection, even when no data is being sent, leading to inefficient resource utilization.
b) It does not support encryption.
c) It requires a new network core, separate from the existing GSM infrastructure.
d) It cannot provide a data rate higher than 9.6 kbps.

82. A radio propagation specialist is analyzing the performance of a UMTS network operating at 2.1 GHz and comparing it with a GSM network operating at 900 MHz. Both networks have the same transmit power and antenna gains. Which network will have a larger cell radius, and why?

a) UMTS, because it uses CDMA which is more robust.
b) UMTS, because it uses a higher frequency which enables better propagation.
c) GSM, because it operates at a lower frequency, which experiences less path loss.
d) Both will have the same cell radius.

83. A mobile station in a GSM network is in idle mode, camping on a cell. It is not actively making a call but must monitor the network for pages (incoming calls) and perform cell reselection as it moves. In GSM, which channel is the mobile station monitoring to receive paging messages?

a) Broadcast Control Channel (BCCH)
b) Traffic Channel (TCH)
c) Common Control Channel (CCCH), specifically the Paging Channel (PCH)
d) Dedicated Control Channel (DCCH)

84. is not available due to congestion, what will be the direct impact on the user?

a) The user will be handed over to a neighboring cell.
b) The user will experience poor voice quality.
c) The user will be able to make calls but not receive them.
d) The user will not be able to make or receive calls, as the signaling procedures cannot be performed.

85. A researcher is studying the evolution of mobile networks from 1G to 5G. The researcher is creating a timeline highlighting the key technological innovations. The researcher correctly notes that 1G used analog FM modulation, 2G used digital modulation (TDMA/FDMA), 3G used CDMA, and 4G used OFDMA. What is the key innovation in 5G that enables the very high data rates and massive connectivity promised by the standard?

a) The elimination of encryption for better performance.
b) The use of millimeter-wave frequencies, massive MIMO, and a flexible numerology.
c) The reuse of the same CDMA technology as 3G.
d) The introduction of a circuit-switched core network.

86. A network operator is configuring a GSM system to minimize interference. The operator is deciding on the frequency reuse factor. A smaller reuse factor (e.g., N=3) gives more channels per cell but increases co-channel interference (CCI). A larger reuse factor (e.g., N=7) reduces CCI but gives fewer channels per cell. The operator wants to maximize the Signal-to-Interference Ratio (SIR) for improved voice quality. Which reuse factor should the operator choose, assuming all other parameters are equal?

a) N=4
b) N=7
c) The reuse factor does not affect SIR.
d) N=3

87. A student of wireless communications is studying the architecture of the GSM network. The student is confused about the difference between the HLR and VLR databases. The HLR is the master database containing all subscriber information, while the VLR is a temporary database for subscribers currently in a particular location. Which of the following correctly describes the relationship between the HLR and VLR during the authentication of a roaming subscriber?

a) The HLR is a subset of the VLR.
b) The VLR stores the subscriber's Ki key.
c) The HLR authenticates the subscriber directly without involving the VLR.
d) The VLR requests authentication triplets (RAND, SRES, Kc) from the HLR for the subscriber.

88. A network planner is designing a network for a new city. The planner wants to use an FDMA system with a total bandwidth of 20 MHz and a channel bandwidth of 200 kHz. The planner must decide between a 7-cell reuse pattern and a 12-cell reuse pattern. The city is expected to have a high density of users. Which reuse pattern will provide the higher number of channels per cell, and what is the consequence for the signal quality?

a) 12-cell pattern gives more channels but higher co-channel interference.
b) 7-cell pattern gives more channels but higher co-channel interference.
c) Both patterns give the same number of channels.
d) 7-cell pattern gives fewer channels but lower co-channel interference.

89. A telecom operator plans to expand its GSM network into rural areas where subscribers frequently travel long distances between adjacent cells. During network testing, engineers observe that calls are maintained even when users move across multiple Base Station Controllers (BSCs). They conclude that the network architecture effectively separates radio resource management from switching operations. Analyze the architecture and determine which GSM component primarily manages radio resources and handovers within its controlled cells.

a) Mobile Switching Centre (MSC)
b) Home Location Register (HLR)
c) Base Station Controller (BSC)
d) Authentication Centre (AuC)

90. A mobile subscriber powers on a GSM phone after arriving in another city. Before allowing incoming or outgoing calls, the network updates the subscriber's temporary location information. Engineers note that this process reduces unnecessary signaling with the Home Location Register while enabling efficient mobility management. Analyze the scenario and identify the GSM database responsible for temporarily storing subscriber information in the visited region.

a) Visitor Location Register (VLR)
b) Base Station Controller (BSC)
c) Equipment Identity Register (EIR)
d) Authentication Centre (AuC)

91. A service provider receives complaints that some stolen mobile phones continue to access the network despite valid subscriber authentication. The engineers decide to verify whether the equipment itself is authorized before permitting network access. Analyze the GSM security architecture and determine which database should be consulted to identify blacklisted mobile equipment.

a) HLR
b) VLR
c) Equipment Identity Register (EIR)
d) AuC

92. A GSM network engineer analyzes the radio interface during periods of heavy traffic. Measurements indicate that the air interface successfully supports bidirectional communication between the Mobile Station and the Base Transceiver Station using standardized signaling procedures. The engineer concludes that a specific GSM interface is responsible for this communication. Analyze the scenario and identify the correct interface.

a) Abis Interface
b) MAP Interface
c) Um Interface
d) A Interface

93. A customer begins a voice call while traveling on a train. As the train moves rapidly through several cellular regions, the network continuously measures signal strength from neighboring cells before deciding whether to transfer the ongoing call. Engineers observe that unnecessary handovers are avoided through predefined thresholds. Analyze the purpose of this process and determine the primary objective of signal measurement during mobility.

a) Reduce subscriber authentication time
b) Encrypt user data packets
c) Increase SMS capacity
d) Ensure seamless communication by selecting the most suitable serving cell

94. A telecom company compares two data services for supporting mobile Internet access. One service allocates a dedicated circuit throughout the communication session, while the other dynamically shares network resources based on packet transmission. Engineers conclude that the packet-based solution offers better bandwidth utilization for bursty Internet traffic. Analyze the technologies and identify the packet-oriented service.

a) Circuit Switching
b) GPRS
c) HSCSD
d) PSTN

95. A financial institution uses a mobile application requiring frequent transmission of small packets containing account updates. During performance evaluation, the network team observes that users pay only for the amount of data transmitted instead of connection duration. Analyze the characteristics of the mobile service and determine which technology best supports this billing model.

a) HSCSD
b) GSM Voice Service
c) GPRS
d) DECT

96. A manufacturing company deploys cordless communication inside its factory campus. The communication system provides high-quality voice services over a limited geographical area without relying on a wide-area cellular infrastructure. Engineers determine that the technology is intended primarily for indoor wireless communication. Analyze the scenario and identify the appropriate technology.

a) DECT
b) UMTS
c) GPRS
d) LTE

97. A university campus plans to replace its wired office telephones with a wireless solution offering mobility within buildings while maintaining excellent voice quality. Since communication is restricted to the campus, wide-area cellular coverage is unnecessary. Analyze the available technologies and determine the most suitable choice.

a) Satellite Communication
b) GSM
c) UMTS
d) DECT

98. A telecom provider upgrades its existing GSM infrastructure to support higher multimedia data rates, video calling, and improved Internet connectivity. Engineers recommend adopting a third-generation mobile technology that employs Wideband Code Division Multiple Access (WCDMA). Analyze the proposed solution and identify the corresponding mobile communication standard.

a) UMTS
b) DECT
c) HSCSD
d) GPRS

99. A mobile application requires real-time video conferencing with simultaneous voice and high-speed data transmission while users travel across cities. Engineers compare second-generation and third-generation technologies before selecting the most suitable solution. Analyze the requirements and identify the technology that best satisfies the application's communication needs.

a) HSCSD
b) GSM
c) UMTS
d) FDMA

100. A mobile network engineer is tasked with optimizing the handover process in a GSM network. The engineer notes that there are two types of handover: hard handover and soft handover. In a hard handover, the connection to the old cell is broken before establishing a connection to the new cell, while in a soft handover, the connection to the new cell is established before breaking the old connection. Which of the following statements best describes a scenario where a soft handover would be beneficial?

a) When the mobile station is in a rural area with large distances between cells.

b) When the network is experiencing low traffic and fewer users are connected.

c) When the mobile station is moving at high speeds between cells.

d) When the mobile station is stationary and connected to a single cell.

